High Performance Parallel Computing
Homework 2

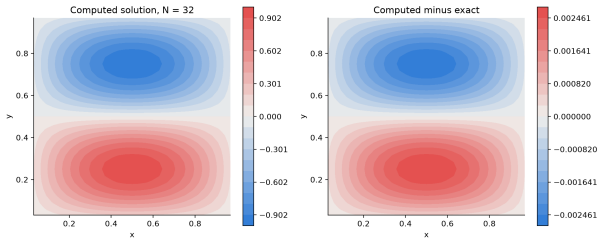
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October 13, 2026

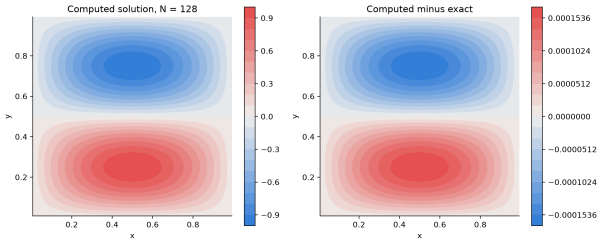
1 Problem 1

1.1 Verification

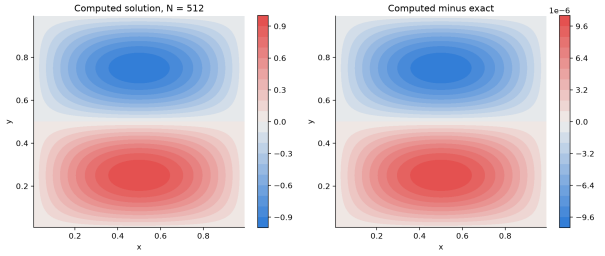
As demonstrated with Figures 1 and 2, the implemeted algorithm converges to the correct solution with the expected quadratic convergence rate.



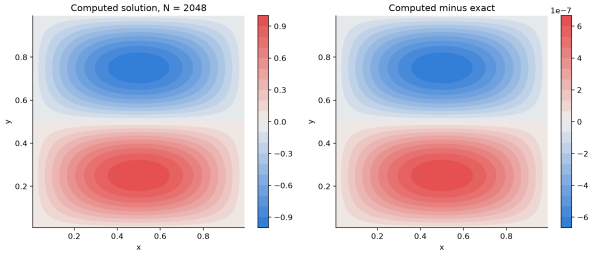
(a) $N = 32$



(b) $N = 128$



(c) $N = 512$



(d) $N = 2048$

Figure 1: Computed solution (left panel of each pair) and the difference between the computed and exact solutions (right panel) for each grid size $N = N_x = N_y$.

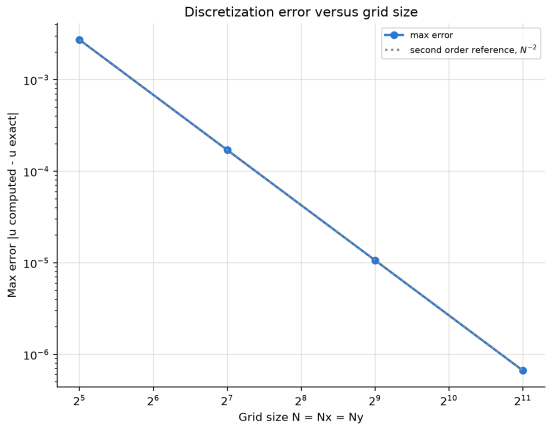
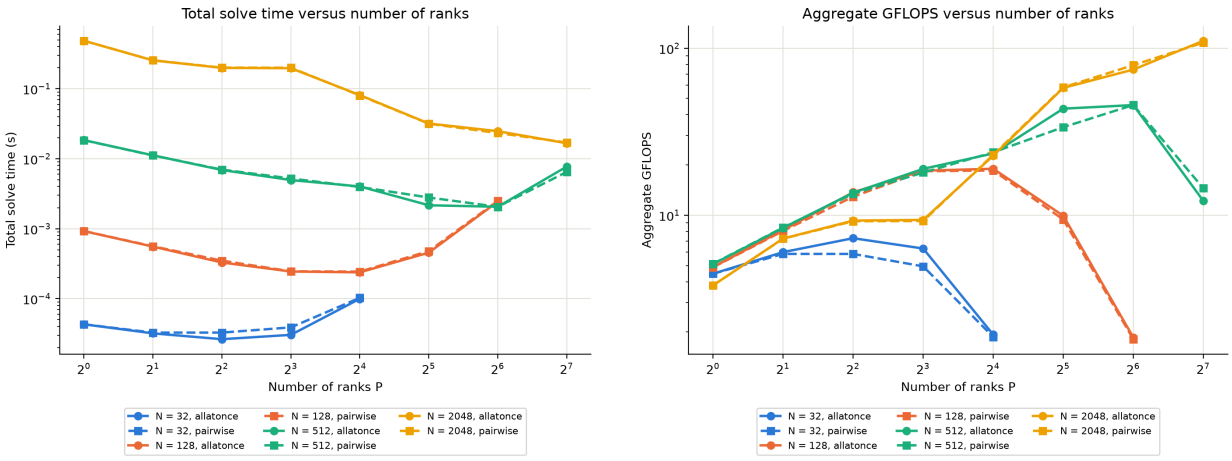


Figure 2: Maximum error of the computed solution versus grid size $N = N_x = N_y$, with an N^{-2} reference line.

1.2 Parallel Scalability

As indicated by Figure 3, the benefits of increased number of ranks in a simulation is very much a function of the size of the problem (N, in this case). For example, as indicated by Figure 3a, the maximum compute time for a single problem size is *maximized* when $P = N/2$, which is in some ways counterintuitive, as one would normally expect the compute time to *strictly decrease* with the number of processors. Interestingly, while the gains in GFLOPS achieved by the increase in processors is clear, the decrease in compute time as a function of processors is marginal at best, given that the total solve time goes from less than a second to less than less than a second[1]. The above results would indicate that gains from parallelism would perhaps be better seen by increasing the problem size further...



(a) Total solve time

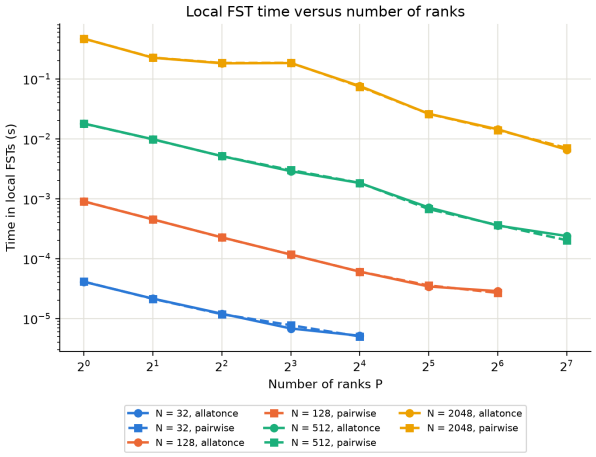
(b) Aggregate GFLOPS

Figure 3: Total solve time and aggregate GFLOPS versus the number of ranks P for each grid size N and both transpose strategies (solid: all-at-once, dashed: pairwise).

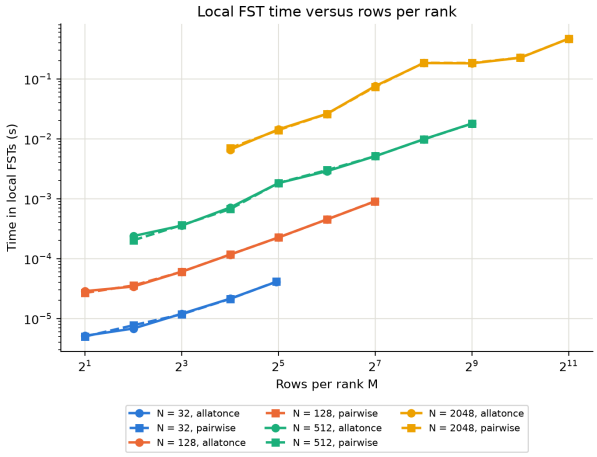
[1]This is indeed stated with a bit of tongue in cheek.

1.3 Transpose Comparison

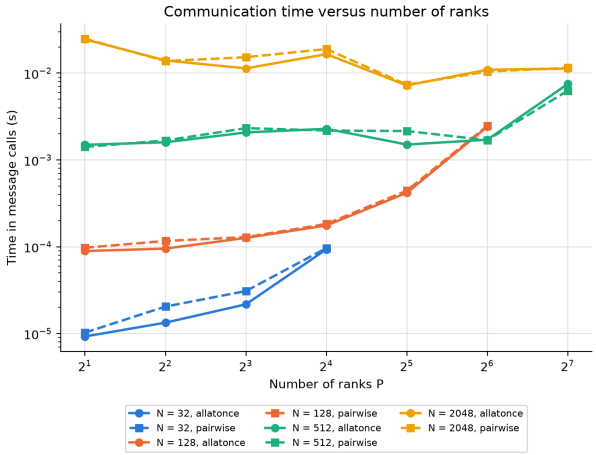
In Figure 4 we compare the communication of the two transpose methods across four different values of n. In the plot Figure 4a shows a largely negligible difference in the two algorithms. The differences in there performances there vary far less than how each method scales with ranks. Figure 4b tells a similar story. The plot which shows to a greater degree how these two differ is Figure 4c. Here we see the all-at-once largely outperform the the pairwise case. Notably, the gap is largest at small n where the latency dominates. It narrows at larger n where this is not the case.



(a) Local FST time versus P



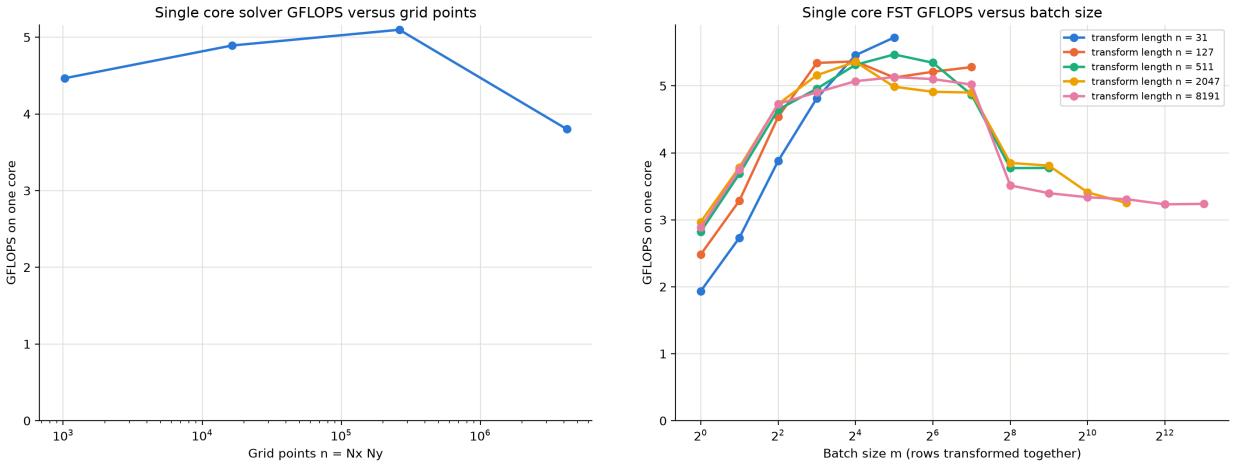
(b) Local FST time versus M



(c) Communication time versus P

Figure 4: Time spent in the local fast sine transforms, versus the number of ranks P and versus the rows per rank M, and time spent in message calls versus P, for each grid size N and both transpose strategies.

1.4 Single Core



(a) Full solve versus grid points $n = N_x N_y$

(b) Batched FST versus batch size m

Figure 5: Single-core GFLOPS of the full Poisson solve and of the batched fast sine transform alone, the latter for each transform length n.

A ping-pong test was performed, with results reported in Figure 6. As expected, the internode message hovers about the same distance away from the intranode messaging, showing how the message communication asymptotically approaches a fixed value of time for jumping a message across nodes.

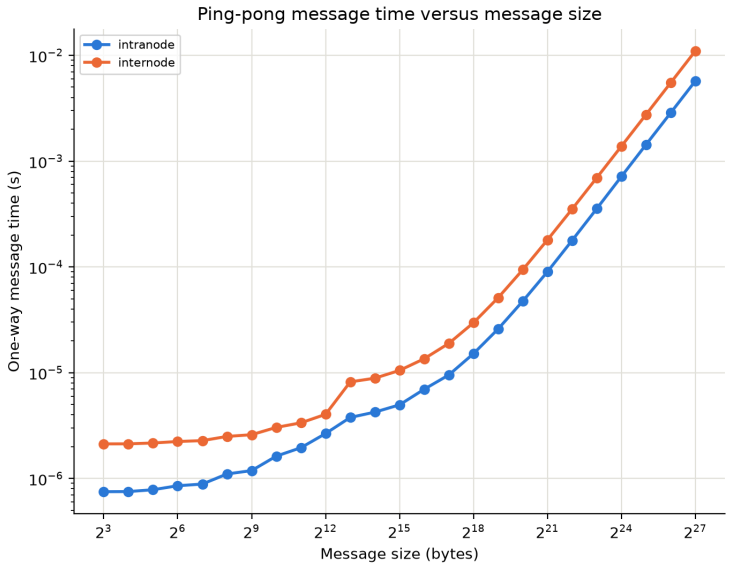


Figure 6: One-way point-to-point message time versus message size from the ping-pong test, with both ranks on one node (intranode) and on two different nodes (internode).

The compute time per rank for P total ranks is:

$$T_P = \text{computer time per rank} + \text{communication} \tag{1}$$

with the ideal being:

$$T_{P,\text{ideal}} = \frac{T_1}{P}. \tag{2}$$

Equation (2) shows that if everything goes our way, we should see a drop off in FST compute time by a factor of two with every doubling of the number of processors. While this is evidently true for $N = 32,128$, the same can certainly *not* be said for $N = 2048$ (see Figure 8). Indeed, for $N = 2048$, we observe a steep drop off in efficiency (notably the number of GFLOPS) as the number of processors increases even though the work required per processor continues to decrease. A possible explanation for this is that the memory required for each processor exceeds the private local cache, meaning memory resources are being accessed from a shared cache, leading to memory contention. This would explain why we eventually see a "bump" back up in efficiency – the required memory is now small enough to be localized, removing the resource fight.

As indicated by Table 4, none of the values passed for N yielded exactly 80% efficiency. However, values for these have been interpolated between neighboring points and reported. Performance limitations are dependent on multiple factors, but notably the problem size. For $N = 32,128$, performance is limited simply by the communication overhead. However, for $N = 2048$, performance could limited by memory access since we are no longer able to store the necessary data for processes in a rank in a local private cache until the number of processors is large enough to fit the split data into small enough chunks.

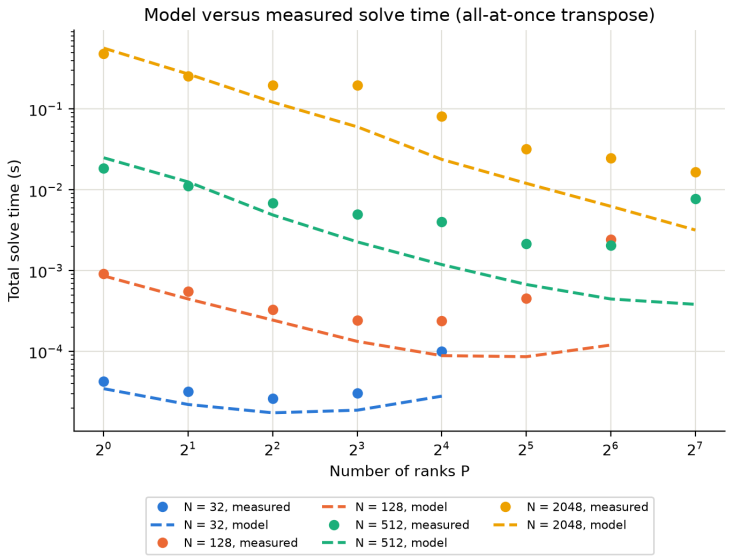


Figure 7: Measured total solve time (markers) and the performance model (dashed lines) versus the number of ranks P for each grid size N, all-at-once transpose.

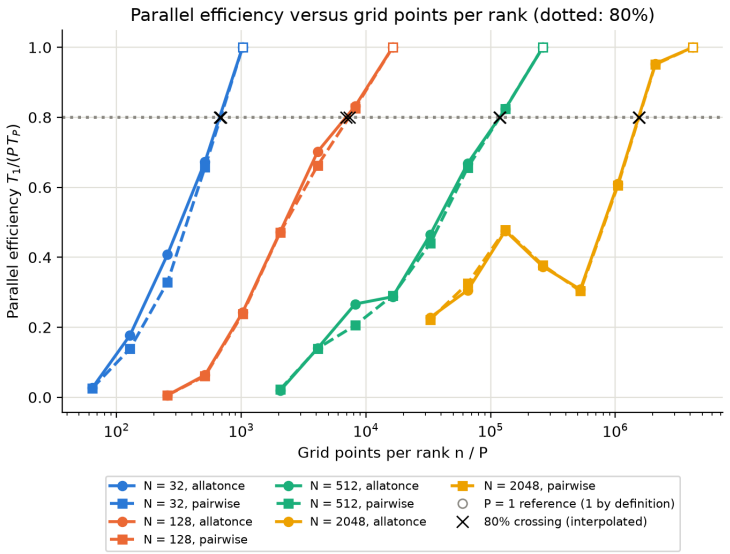


Figure 8: Parallel efficiency $T_1/(P\,T_P)$ versus grid points per rank n/P for each grid size N and both transpose strategies. The dotted line marks 80% efficiency. Hollow markers are the $P = 1$ reference runs, whose efficiency is 1 by definition; crosses mark where each curve passes through 80%.

P	Nx	Ny	mode	M	t_total	t_fst	t_transpose	t_comm	gflops	error
1	128	128	allatonce	127	9.23E-04	9.00E-04	2.29E-05	1.60E-07	4.89	1.71E-04
1	128	128	pairwise	127	9.27E-04	9.03E-04	2.30E-05	0.00E+00	4.87	1.71E-04
1	2048	2048	allatonce	2047	4.85E-01	4.69E-01	1.63E-02	8.01E-07	3.80	6.67E-07
1	2048	2048	pairwise	2047	4.85E-01	4.68E-01	1.63E-02	0.00E+00	3.80	6.67E-07
1	32	32	allatonce	31	4.30E-05	4.10E-05	1.81E-06	1.30E-07	4.47	2.73E-03
1	32	32	pairwise	31	4.31E-05	4.13E-05	1.58E-06	0.00E+00	4.46	2.73E-03
1	512	512	allatonce	511	1.84E-02	1.80E-02	4.85E-04	1.80E-07	5.10	1.07E-05
1	512	512	pairwise	511	1.84E-02	1.79E-02	4.85E-04	0.00E+00	5.12	1.07E-05
2	128	128	allatonce	64	5.55E-04	4.50E-04	1.07E-04	8.93E-05	8.14	1.71E-04
2	128	128	pairwise	64	5.61E-04	4.51E-04	1.14E-04	9.74E-05	8.05	1.71E-04
2	2048	2048	allatonce	1024	2.54E-01	2.25E-01	3.20E-02	2.44E-02	7.25	6.67E-07
2	2048	2048	pairwise	1024	2.55E-01	2.27E-01	3.20E-02	2.49E-02	7.24	6.67E-07
2	32	32	allatonce	16	3.20E-05	2.15E-05	1.03E-05	9.27E-06	6.01	2.73E-03
2	32	32	pairwise	16	3.28E-05	2.13E-05	1.13E-05	1.03E-05	5.86	2.73E-03
2	512	512	allatonce	256	1.12E-02	9.82E-03	1.77E-03	1.49E-03	8.40	1.07E-05
2	512	512	pairwise	256	1.11E-02	9.77E-03	1.72E-03	1.41E-03	8.44	1.07E-05
4	128	128	allatonce	32	3.29E-04	2.26E-04	1.05E-04	9.53E-05	13.73	1.71E-04
4	128	128	pairwise	32	3.50E-04	2.25E-04	1.25E-04	1.17E-04	12.91	1.71E-04
4	2048	2048	allatonce	512	1.99E-01	1.82E-01	1.89E-02	1.40E-02	9.28	6.67E-07
4	2048	2048	pairwise	512	2.00E-01	1.84E-01	2.10E-02	1.38E-02	9.20	6.67E-07
4	32	32	allatonce	8	2.64E-05	1.20E-05	1.44E-05	1.34E-05	7.28	2.73E-03
4	32	32	pairwise	8	3.28E-05	1.18E-05	2.11E-05	2.05E-05	5.86	2.73E-03
4	512	512	allatonce	128	6.90E-03	5.12E-03	1.81E-03	1.60E-03	13.63	1.07E-05
4	512	512	pairwise	128	7.00E-03	5.13E-03	1.92E-03	1.68E-03	13.44	1.07E-05
8	128	128	allatonce	16	2.44E-04	1.17E-04	1.31E-04	1.27E-04	18.48	1.71E-04
8	128	128	pairwise	16	2.46E-04	1.17E-04	1.33E-04	1.29E-04	18.34	1.71E-04
8	2048	2048	allatonce	256	1.97E-01	1.84E-01	1.59E-02	1.13E-02	9.38	6.67E-07
8	2048	2048	pairwise	256	2.00E-01	1.85E-01	1.91E-02	1.53E-02	9.23	6.67E-07
8	32	32	allatonce	4	3.04E-05	6.82E-06	2.37E-05	2.17E-05	6.32	2.73E-03
8	32	32	pairwise	4	3.89E-05	7.74E-06	3.17E-05	3.09E-05	4.94	2.73E-03
8	512	512	allatonce	64	4.96E-03	2.88E-03	2.16E-03	2.07E-03	18.95	1.07E-05
8	512	512	pairwise	64	5.22E-03	3.01E-03	2.42E-03	2.32E-03	18.01	1.07E-05
16	128	128	allatonce	8	2.38E-04	6.06E-05	1.78E-04	1.76E-04	18.97	1.71E-04
16	128	128	pairwise	8	2.44E-04	6.05E-05	1.86E-04	1.83E-04	18.53	1.71E-04
16	2048	2048	allatonce	128	8.12E-02	7.59E-02	1.88E-02	1.65E-02	22.72	6.67E-07
16	2048	2048	pairwise	128	8.02E-02	7.40E-02	2.09E-02	1.89E-02	23.00	6.67E-07
16	32	32	allatonce	2	9.98E-05	5.16E-06	9.45E-05	9.34E-05	1.93	2.73E-03
16	32	32	pairwise	2	1.03E-04	5.02E-06	9.80E-05	9.66E-05	1.87	2.73E-03
16	512	512	allatonce	32	4.01E-03	1.82E-03	2.35E-03	2.28E-03	23.43	1.07E-05
16	512	512	pairwise	32	3.95E-03	1.84E-03	2.22E-03	2.18E-03	23.81	1.07E-05
32	128	128	allatonce	4	4.55E-04	3.42E-05	4.22E-04	4.18E-04	9.93	1.71E-04
32	128	128	pairwise	4	4.79E-04	3.57E-05	4.46E-04	4.43E-04	9.43	1.71E-04
32	2048	2048	allatonce	64	3.19E-02	2.61E-02	8.87E-03	7.21E-03	57.86	6.67E-07
32	2048	2048	pairwise	64	3.16E-02	2.60E-02	8.39E-03	7.36E-03	58.27	6.67E-07
32	512	512	allatonce	16	2.16E-03	7.14E-04	1.52E-03	1.50E-03	43.46	1.07E-05
32	512	512	pairwise	16	2.79E-03	6.70E-04	2.17E-03	2.15E-03	33.66	1.07E-05
64	128	128	allatonce	2	2.44E-03	2.87E-05	2.41E-03	2.41E-03	1.85	1.71E-04
64	128	128	pairwise	2	2.50E-03	2.68E-05	2.47E-03	2.47E-03	1.81	1.71E-04
64	2048	2048	allatonce	32	2.48E-02	1.44E-02	1.17E-02	1.09E-02	74.48	6.67E-07
64	2048	2048	pairwise	32	2.33E-02	1.40E-02	1.06E-02	1.04E-02	79.16	6.67E-07
64	512	512	allatonce	8	2.06E-03	3.56E-04	1.73E-03	1.71E-03	45.65	1.07E-05
64	512	512	pairwise	8	2.06E-03	3.63E-04	1.71E-03	1.69E-03	45.69	1.07E-05
128	2048	2048	allatonce	16	1.66E-02	6.57E-03	1.14E-02	1.13E-02	110.81	6.67E-07
128	2048	2048	pairwise	16	1.71E-02	7.02E-03	1.16E-02	1.15E-02	108.05	6.67E-07
128	512	512	allatonce	4	7.72E-03	2.38E-04	7.50E-03	7.49E-03	12.18	1.07E-05
128	512	512	pairwise	4	6.49E-03	2.03E-04	6.29E-03	6.28E-03	14.49	1.07E-05

Table 1: Timings for FST and transpositions. Also reporting alongside the number of GFLOPS and the error.

Table 2: tab:timings

P	Nx	Ny	mode	M	FST %	Transpose %	Communication %	gflops
1	128	128	allatonce	127	97.48%	2.48%	0.02%	4.89
1	128	128	pairwise	127	97.49%	2.48%	0.00%	4.87
1	2048	2048	allatonce	2047	96.64%	3.35%	0.00%	3.80
1	2048	2048	pairwise	2047	96.65%	3.35%	0.00%	3.80
1	32	32	allatonce	31	95.25%	4.21%	0.30%	4.47
1	32	32	pairwise	31	95.79%	3.67%	0.00%	4.46
1	512	512	allatonce	511	97.37%	2.63%	0.00%	5.10
1	512	512	pairwise	511	97.36%	2.64%	0.00%	5.12
2	128	128	allatonce	64	81.13%	19.34%	16.09%	8.14
2	128	128	pairwise	64	80.49%	20.33%	17.37%	8.05
2	2048	2048	allatonce	1024	88.47%	12.59%	9.61%	7.25
2	2048	2048	pairwise	1024	88.91%	12.58%	9.78%	7.24
2	32	32	allatonce	16	67.22%	32.06%	28.99%	6.01
2	32	32	pairwise	16	65.00%	34.33%	31.37%	5.86
2	512	512	allatonce	256	87.72%	15.80%	13.34%	8.40
2	512	512	pairwise	256	87.72%	15.41%	12.69%	8.44
4	128	128	allatonce	32	68.76%	31.91%	28.96%	13.73
4	128	128	pairwise	32	64.35%	35.61%	33.36%	12.91
4	2048	2048	allatonce	512	91.49%	9.50%	7.05%	9.28
4	2048	2048	pairwise	512	91.99%	10.47%	6.86%	9.20
4	32	32	allatonce	8	45.59%	54.56%	50.64%	7.28
4	32	32	pairwise	8	36.00%	64.36%	62.47%	5.86
4	512	512	allatonce	128	74.31%	26.24%	23.16%	13.63
4	512	512	pairwise	128	73.36%	27.41%	23.95%	13.44
8	128	128	allatonce	16	47.77%	53.50%	51.78%	18.48
8	128	128	pairwise	16	47.64%	53.84%	52.41%	18.34
8	2048	2048	allatonce	256	93.43%	8.08%	5.76%	9.38
8	2048	2048	pairwise	256	92.71%	9.57%	7.64%	9.23
8	32	32	allatonce	4	22.42%	77.94%	71.45%	6.32
8	32	32	pairwise	4	19.88%	81.43%	79.45%	4.94
8	512	512	allatonce	64	57.98%	43.61%	41.80%	18.95
8	512	512	pairwise	64	57.58%	46.30%	44.51%	18.01
16	128	128	allatonce	8	25.46%	74.80%	73.83%	18.97
16	128	128	pairwise	8	24.82%	76.11%	75.11%	18.53
16	2048	2048	allatonce	128	93.56%	23.21%	20.37%	22.72
16	2048	2048	pairwise	128	92.38%	26.01%	23.61%	23.00
16	32	32	allatonce	2	5.17%	94.75%	93.60%	1.93
16	32	32	pairwise	2	4.87%	95.18%	93.77%	1.87
16	512	512	allatonce	32	45.45%	58.58%	56.89%	23.43
16	512	512	pairwise	32	46.48%	56.27%	55.24%	23.81
32	128	128	allatonce	4	7.52%	92.67%	91.90%	9.93
32	128	128	pairwise	4	7.45%	93.12%	92.48%	9.43
32	2048	2048	allatonce	64	81.99%	27.82%	22.63%	57.86
32	2048	2048	pairwise	64	82.23%	26.52%	23.26%	58.27
32	512	512	allatonce	16	32.99%	70.37%	69.53%	43.46
32	512	512	pairwise	16	24.01%	77.58%	77.03%	33.66
64	128	128	allatonce	2	1.18%	98.97%	98.88%	1.85
64	128	128	pairwise	2	1.07%	99.05%	98.83%	1.81
64	2048	2048	allatonce	32	58.15%	47.33%	44.18%	74.48
64	2048	2048	pairwise	32	60.12%	45.51%	44.70%	79.16
64	512	512	allatonce	8	17.29%	83.91%	83.04%	45.65
64	512	512	pairwise	8	17.63%	82.89%	82.28%	45.69
128	2048	2048	allatonce	16	39.51%	68.50%	67.70%	110.81
128	2048	2048	pairwise	16	41.15%	67.81%	67.35%	108.05
128	512	512	allatonce	4	3.09%	97.16%	97.03%	12.18
128	512	512	pairwise	4	3.13%	97.00%	96.78%	14.49

Table 3: Percentage of total time spent for solving via FST, data transposition, and communication.

N	transpose	bracketing P	$P_{0.8}$	$n_{0.8} = n/P_{0.8}$	GFLOPS
32	allatonce	1 and 2	1.53	670	5.46
32	pairwise	1 and 2	1.5	683	5.35
128	allatonce	2 and 4	2.37	6.92E+03	9.27
128	pairwise	2 and 4	2.23	7.33E+03	8.71
512	allatonce	2 and 4	2.22	1.18E+05	9.06
512	pairwise	2 and 4	2.21	1.18E+05	9.06
2048	allatonce	2 and 4	2.73	1.54E+06	8.3
2048	pairwise	2 and 4	2.71	1.55E+06	8.24

Table 4: Rank count $P_{0.8}$, grid points per rank $n_{0.8}$ and aggregate GFLOPS at 80% parallel efficiency. Each curve is interpolated linearly in $\log_2 P$ between the two measured rank counts that bracket 80%, and the GFLOPS value is $0.8\,P_{0.8}$ times the single-rank GFLOPS.